SUBFUNCTORS OF ALGEBRAIC SPACES SATISFYING THE AXIOMS ARE ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2683-2695 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 2697-2708, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 44 of queue_artin.json (pb=568, stmt 2683-2695, proof 2697-2708).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07X1: sources/stacks-artin/pinned-original-source.tex, source lines 1030–1037. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07Y1: sources/stacks-artin/pinned-original-source.tex, source lines 2641–2649. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $a : F \to G$ be a transformation of functors $(Sch/S)^{opp}_{fppf} \to Sets$. Assume that*

1. *a is injective,*
2. *F satisfies axioms [0], [1], [2], [4], and [5],*
3. *$\mathcal{O}_{S,s}$ is a G-ring for all finite type points s of S,*
4. *G is an algebraic space locally of finite type over S,*

Then F is an algebraic space.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By Lemma 07X1 the functor G satisfies [3]. As $F \to G$ is injective, we conclude that F also satisfies [3]. Moreover, as $F \to G$ is injective, we see that given schemes U, V and morphisms $U \to F$ and $V \to F$, then $U \times_F V = U \times_G V$. Hence $\Delta : F \to F \times F$ is representable (by schemes) as this holds for G by assumption. Thus Proposition 07Y1 applies[1]. □

[1]The set theoretic condition [-1] holds for F as it holds for G. Details omitted.